The Nile at NADI 2026 Subtask 1.1: A Two-Phase Acoustic Fine-Tuning Strategy for Robust Dialectal Arabic Speech Recognition

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Abstract

We describe our submission to **NADI 2026 Subtask 1.1 (Noisy Country-Level ASR)** (Sullivan et al., 2026), a two-phase acoustic fine-tuning recipe on WHISPER-LARGE-V3-TURBO. *Phase 1* adapts the model to ∼100k dialectal utterances (∼496 h) under stochastic degradation (noise, bandwidth, codec, gain, speed) applied to otherwise clean audio. *Phase 2* anneals on the 22,608-utterance in-domain NADI target at a reduced learning rate. On the official 4,000-utterance blind test set we reach **57.0 WER** (6th of 9); compared with a single-phase model trained on the identical pooled data, the two-phase schedule yields an **8.0 absolute WER point** improvement.

1 Introduction

Whisper (Radford et al., 2023) achieves strong zero-shot performance on Modern Standard Arabic but degrades markedly on Dialectal Arabic (DA) (Hussein et al., 2022; Chowdhury et al., 2021), whose Egyptian, Levantine, Gulf, Iraqi, and Maghrebi varieties differ substantially in phonology and lexicon (Shon et al., 2020). Real-world DA audio adds narrowband encoding, codec artifacts, and background noise—the regime targeted by **NADI 2026 Subtask 1.1** (Sullivan et al., 2026), which ranks systems by WER on utterances from eight countries under degraded conditions.

Naïvely fine-tuning a large model on one dialectal corpus overfits its acoustics without expanding phonetic coverage; conversely, fine-tuning purely on augmented multi-dialect data misaligns outputs from the evaluation distribution (punctuation, orthography, target vocabulary). We propose a **two-phase acoustic fine-tuning** recipe that separates these objectives: **Phase 1** adapts to broad dialects and acoustic conditions using clean multi-dialect audio under heavy *synthetic* degradation, exposing the model to a diverse span of channel and noise conditions; **Phase 2** anneals on the naturally noisy, in-domain NADI target at a lower learning rate, aligning outputs to the *real* target noise and orthographic distributions. Our contributions are (i) a competitive DA-ASR pipeline on Whisper-Large-v3-Turbo (6th/9 on NADI 2026 Subtask 1.1), (ii) a simple orthogonal degradation pipeline, and (iii) a controlled comparison showing an 8.0 WER-point improvement of the two-phase schedule over a single-phase pooled-data baseline.

2 Related Work

Recent multilingual encoder-decoder speech models—Whisper (Radford et al., 2023), USM (Zhang et al., 2023), SeamlessM4T (Barrault et al., 2023), MMS (Pratap et al., 2024)—are standard starting points for low-resource ASR fine-tuning; Whisper's LARGE-V3-TURBO halves parameter count by pruning the decoder from 32 to 4 layers while keeping the full 32-layer encoder. Self-supervised alternatives include wav2vec 2.0 (Baevski et al., 2020), HuBERT (Hsu et al., 2021), XLS-R (Babu et al., 2022), and Conformer encoders (Gulati et al., 2020); see Prabhavalkar et al. (2024) for a survey. Arabic-specific work has progressed via end-to-end and modular systems (Hussein et al., 2022) and unified multilingual code-switching strategies (Chowdhury et al., 2021), with resources like ADI17 (Shon et al., 2020), Common Voice (Ardila et al., 2020), and FLEURS (Conneau et al., 2023) broadening evaluation. Our submission targets NADI 2026 (Sullivan et al., 2026), whose official baseline is zero-shot Whisper-Large-

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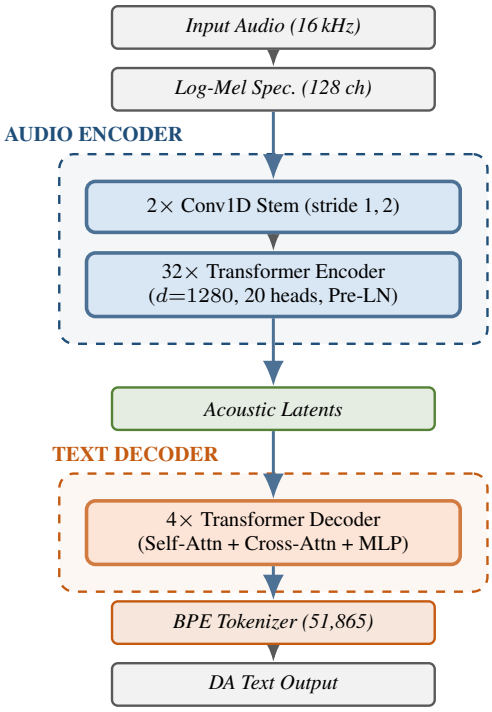


Figure 1: WHISPER-LARGE-V3-TURBO: 32-layer encoder retained, decoder pruned to 4 layers.

v3.

3 Method

Base model. We build on WHISPER-LARGE-V3-TURBO (Radford et al., 2023) (∼809M params). Audio is resampled to 16 kHz mono and converted to a **128-channel** log-Mel spectrogram (25 ms window, 10 ms hop). A two-layer 1D convolutional stem (kernel 3, strides 1 and 2) downsamples the time dimension by a **factor of 2** and projects into $d_{\text{model}} = 1280$. The 32-layer encoder uses Pre-LayerNorm and 20-head self-attention; the 4-layer decoder attends autoregressively over a 51,865-token BPE vocabulary. Figure 1 summarizes the architecture.

Two-phase fine-tuning. Phase 1 fine-tunes on a ∼100k-utterance subset of **ODAMIX** (`dialectal-arabic-lahgtna-v2` (oddadmix, 2026)), a multi-dialect corpus spanning 13 country-level varieties (Maghrebi 33.5%, Egyptian 27.1%, Gulf 14.9%, Levantine 10.9%, Sudanese/Yemeni/Iraqi 13.7%; no MSA). ODAMIX audio is comparatively clean; each mini-batch is passed through the stochastic degradation pipeline of Table 1 to expose the encoder to a synthetic diversity of channel and noise conditions. **Phase 2** anneals the resulting checkpoint on the NADI 2026 target at 2×10^{-5}. Unlike Phase 1, the NADI Subtask 1.1 audio is naturally noisy—the subtask is titled "Noisy Country-Level ASR"—so no additional synthetic degradation is applied here; the reduced step size lets the model align to the target's real noise distribution and output conventions (punctuation, orthography, vocabulary) while limiting catastrophic forgetting of dialectal representations from Phase 1. Figure 2 summarizes the pipeline.

Transform	Exact setting	P
White noise	Gaussian, SNR 3–15 dB	0.10
Pink noise	$1/f$ spectrum, SNR 3–15 dB	0.10
Bandwidth cut	8 kHz decimation → 16 kHz	0.10
Codec sim.	GSM 06.10 full-rate, 13 kbps	0.10
Gain shift	attenuation, −15 to 0 dB	0.10
Speed perturb.	resample, $\{0.9, 1.0, 1.1\}$	0.10

Table 1: Phase-1 stochastic degradation pipeline (independent per example).

Augmentation pipeline. Six transformations are applied independently at P=0.10 each (Table 1), yielding $1-0.9^6 \approx 46.9\%$ of examples receiving at least one perturbation. Ranges target orthogonal failure modes—additive noise, narrowband telephony, codec artifacts, quiet/distant speakers, and varied speaking rates—without inducing phonetic drift.

4 Experimental Setup

Data. NADI 2026 Subtask 1.1 (Sullivan et al., 2026) releases 12,800 train + 10,808 validation + 4,000 blind-test utterances over eight countries (Algeria, Egypt, Jordan, Mauritania, Morocco, Palestine, UAE, Yemen). We train on 22,608 (train + 9,808 val) and hold out 1,000 val utterances as internal dev. Phase-1 ODAMIX is a ∼100k subset (≈496 h) of the 611,180-utterance `oddadmix/dialectal-arabic-lahgtna-v2` (oddadmix, 2026); the corpus carries no speaker annotations.

Training and decoding. Both phases perform **full fine-tuning** (no frozen layers, no LoRA) with AdamW (β_1=0.9, β_2=0.999, ϵ=10^{-8}) on Hugging Face Transformers (Wolf et al., 2020). Table 2 summarizes hyperparameters. Phase 1 ran on 1×A100 40 GB (≈6 h); Phase 2 on 1×RTX PRO 6000 Blackwell (1 h 20 m, 1,416 steps) with bf16 autocast + fp32 master weights. Decoding uses **greedy search** (beam 1),



Figure 2: Two-phase fine-tuning pipeline. Phase 1 broadens dialectal and acoustic coverage under stochastic degradation on ∼100k ODAMIX utterances; the checkpoint is annealed on the NADI 2026 target at a reduced LR in Phase 2.

Parameter	Phase 1	Phase 2
Init. weights	Turbo	Phase-1 ckpt
Dataset	ODAMIX (∼100k)	NADI (22,608)
Epochs	3	4
Learning rate	5×10^{-5}	2×10^{-5}
Batch size	32	32
Warm-up	not logged	300 steps
Weight decay	not logged	0.0
Seed	not logged	42
Precision	not logged	bf16 / fp32
GPU	1×A100 40 GB	1×RTX 6000
Train time	≈6 h	1 h 20 m
Max audio / tok	30 s / 448	30 s / 448
Augmentation	P=0.10 each	none

Table 2: Hyperparameters. Both phases fine-tune all parameters, greedy decoding. "Not logged" indicates hyperparameters we did not record for the Phase-1 run.

`language="ar"`, `task="transcribe"`, max 225 output tokens.

Evaluation. We report WER on our 1,000-utterance internal dev (in-loop scoring: diacritic + tatweel stripping) and the official 4,000-utterance test set. **All test-set WER values reported in Table 3 are computed with the official organizers' scorer** (which additionally normalizes digits, punctuation, and Alif/Ya/Ta-Marbouta variants). Dev figures are therefore slightly pessimistic relative to the official protocol.

5 Results

Table 3 reports WER. Zero-shot Whisper-Turbo reaches 80.0; direct target fine-tuning reduces to 71.5; Phase 1 alone reaches 65.0, confirming that broad multi-dialect exposure under degradation transfers even without target supervision. The full two-phase pipeline reaches **57.0**—a 23.0-point absolute (28.8% relative) reduction over zero-shot, **ranking 6th of 9 submitted systems**.

Two-phase vs. single-phase pooled baseline. Pooling the identical Phase-1 and target corpora into a single training run reaches only 65.0—matching Phase 1 alone—while the two-phase split reaches 57.0. *Compared with the single-phase pooled-data baseline, the two-phase schedule improves WER by 8.0 absolute points (12.3% relative)*, holding the training corpora fixed. We refrain from a stronger causal claim ("schedule alone") because the two configurations differ modestly in effective LR schedule and total step count (Phase 1: 3 epochs at 5×10^{-5} on ∼100k utts, then Phase 2: 4 epochs at 2×10^{-5} on 22.6k utts; single-phase pooled: comparable epochs at a single LR on the union). We interpret the finding as: given a fixed corpus, splitting fine-tuning into a high-LR broad-adaptation stage and a low-LR domain-annealing stage substantially outperforms a single flat schedule.

Why the decomposition helps. A single-phase target fine-tune memorizes target acoustics without learning dialect-general representations; a single-phase ODAMIX fine-tune produces a robust but stylistically misaligned model. The two-phase recipe assigns each objective to the phase best suited to optimize it: broad synthetically-degraded data at higher LR for representation learning, real in-domain data at lower LR for output alignment.

6 Conclusion

We described our NADI 2026 Subtask 1.1 submission: a two-phase acoustic fine-tuning strategy on Whisper-Large-v3-Turbo that separates *dialectal/acoustic robustness* (Phase 1, synthetic degradation on clean multi-dialect data) from *target-domain alignment* (Phase 2, annealing on the naturally noisy in-domain NADI target). The pipeline achieves 57.0 WER on the official blind test set (6th/9), an 8.0-point improvement over a single-phase pooled-data baseline trained on the identical corpora.

Limitations

Several caveats bound our claims. First, we compare the two-phase schedule against single-phase pooled training, but we do not isolate augmentation itself: a Phase 1 on clean ODAMIX *without* synthetic degradation, followed by annealing,

System	Dev	Test	Δ^{a}	Rel.%[b]
Whisper-Turbo (zero-shot)	—	80.0	—	—
+ Phase 2 only (target FT)	—	71.5	−8.5	10.6%
+ Phase 1 only (ODAMIX+aug.)	—	65.0	−15.0	18.8%
+ Phase 1 → Phase 2 (ours, rank 6/9)	**61.4**	**57.0**	**−23.0**	**28.8%**
Ablation — same data, no two-phase schedule (official test)				
Combined single-phase (no annealing)	—	65.0	−15.0	18.8%

Table 3: WER (%) on NADI 2026 Subtask 1.1; all test values computed with the official organizers' scorer. [a]Absolute WER reduction vs. zero-shot. [b]Relative reduction vs. zero-shot. Our two-phase pipeline (highlighted) ranks **6th of 9 submitted systems**. Splitting the same pooled data into two phases yields an **8.0-point improvement** over the single-phase baseline (65.0 → 57.0), a **12.3% relative** reduction under identical training corpora.

would be required to separate the effect of augmentation from that of the additional dialectal data—we leave this to future work. Second, the two-phase vs. single-phase comparison does not perfectly match total step count or LR schedule, so the 8.0-point improvement should be read as a comparison against a natural single-phase baseline rather than a claim about the schedule "alone". Third, only Phase 2 was run with a fixed seed (42); Phase-1 seed, warm-up, precision, and weight decay were not logged, and we report no variance across seeds in either phase. Fourth, ODAMIX has no speaker identifiers, so speaker overlap between phases cannot be audited. Fifth, the Phase 1 corpus is Maghrebi/Egyptian-heavy with no MSA; performance on under-represented dialects (e.g. Iraqi, 2.1% of P1) may be systematically weaker than the aggregate WER suggests. Finally, reverberation and room-impulse-response convolution are not part of our augmentation set; both are natural extensions.

References

Rosana Ardila, Megan Branson, Kelly Davis, Michael Kohler, Josh Meyer, Michael Henretty, Reuben Morais, Lindsay Saunders, Francis Tyers, and Gregor Weber. 2020. Common voice: A massively-multilingual speech corpus. In *Language Resources and Evaluation Conference (LREC)*, pages 4218–4222.

Arun Babu, Changhan Wang, Andros Tjandra, Kushal Lakhotia, Qiantong Xu, Naman Goyal, Kritika Singh, Patrick Von Platen, Yatharth Saraf, Juan Pino, et al. 2022. XLS-R: Self-supervised cross-lingual speech representation learning at scale. In *Interspeech*.

Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. wav2vec 2.0: A framework for self-supervised learning of speech representations. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 12449–12460.

Loïc Barrault, Yu-An Chung, Mariano Cora Meglioli, David Dale, Ning Dong, Paul-Ambroise Duquenne, Hady Elsahar, Hongyu Gong, Kevin Heffernan, John Hoffman, et al. 2023. SeamlessM4T: Massively multilingual and multimodal machine translation. *arXiv preprint arXiv:2308.11596*.

Shammur Absar Chowdhury, Amir Hussein, Ahmed Abdelali, and Ahmed Ali. 2021. Towards one model to rule all: Multilingual strategy for dialectal code-switching Arabic ASR. In *Interspeech*.

Alexis Conneau, Min Ma, Simran Khanuja, Yu Zhang, Vera Axelrod, Siddharth Dalmia, Jason Riesa, Clara Rivera, and Ankur Bapna. 2023. FLEURS: Few-shot learning evaluation of universal representations of speech. In *IEEE Spoken Language Technology Workshop (SLT)*, pages 798–805.

Anmol Gulati, James Qin, Chung-Cheng Chiu, Niki Parmar, Yu Zhang, Jiahui Yu, Wei Han, Shibo Wang, Zhengdong Zhang, Yonghui Wu, et al. 2020. Conformer: Convolution-augmented transformer for speech recognition. In *Interspeech*.

Wei-Ning Hsu, Benjamin Bolte, Yao-Hung Hubert Tsai, Kushal Lakhotia, Ruslan Salakhutdinov, and Abdelrahman Mohamed. 2021. HuBERT: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 29:3451–3460.

Amir Hussein, Shinji Watanabe, and Ahmed Ali. 2022. Arabic speech recognition by end-to-end, modular systems and human. *Computer Speech & Language*, 71:101272.

oddadmix. 2026. Dialectal Arabic lahgtna v2. Hugging Face dataset repository. Accessed 21 August 2026.

Rohit Prabhavalkar, Takaaki Hori, Tara N Sainath, Ralf Schlüter, and Shinji Watanabe. 2024. End-to-end speech recognition: A survey. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 32:325–351.

Vineel Pratap, Andros Tjandra, Bowen Shi, Paden Tomasello, Arun Babu, Sayani Kundu, Ali Elkahky, Zhaoheng Ni, Apoorv Vyas, Maryam Fazel-Zarandi,

et al. 2024. Scaling speech technology to 1,000+ languages. *Journal of Machine Learning Research*, 25(97):1–52.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *International Conference on Machine Learning (ICML)*, pages 28492–28518.

Suwon Shon, Ahmed Ali, Younes Samih, Hamdy Mubarak, and James Glass. 2020. ADI17: A fine-grained Arabic dialect identification dataset. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 8244–8248.

Peter Sullivan, Bashar Talafha, Ahmed Ashraf, Fethi Bougares, Haroun Elleuch, Chiyu Zhang, Abdel-Rahim Elmadany, Youssef Mohamed, Salima Mdhaffar, Yannick Estève, Mohamed Elhoseiny, Hamzah Luqman, Nizar Habash, and Muhammad Abdul-Mageed. 2026. NADI-2026: The second multidialectal Arabic speech processing shared task. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, et al. 2020. Transformers: State-of-the-art natural language processing. In *Conference on Empirical Methods in Natural Language Processing (EMNLP): System Demonstrations*, pages 38–45.

Yu Zhang, Wei Han, James Qin, Yongqiang Wang, Ankur Bapna, Zhehuai Chen, Nanxin Chen, Bo Li, Vera Axelrod, Gary Wang, et al. 2023. Google USM: Scaling automatic speech recognition beyond 100 languages. *arXiv preprint arXiv:2303.01037*.